

Adaptive Neuro-Fuzzy Inference System Identification for the Dynamics of the AR.Drone Quadcopter

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Abstract—In this paper, we study the non-linear modelling of the lateral, longitudinal, and vertical dynamics of the AR.Drone Quadcopter by means of the Adaptive Neuro-Fuzzy Inference Systems (ANFIS) identification technique. We derive the multi-input multi-output (MIMO) ANFIS model of the inner (attitude) loop systems. We employ the Dryden wind turbulence model to represent a realistic wind gust scenario in real flight environments. Furthermore, we benchmark the performance of our proposed ANFIS models with respect to the performance of the linear system identification techniques. This paper serves as a preliminary study towards our long-term goal in developing robust autopilot systems for a quadcopter drone using neuro-fuzzy models, which have multiple advantages over the traditional (mathematical-based) modelling techniques, such as the knowledge-based property of the fuzzy principle, which is not only transparent, but also suitable to accommodate ambiguity; in addition to its adaptive nature. The learning capability of the neural networks is also suitable to represent the dynamics of highly non-linear systems as in the case of our quadcopter drone.

Index Terms—Adaptive Neuro-Fuzzy, System Identification, the AR.Drone Quadcopter.

I. INTRODUCTION

Robotic aircraft have been widely implemented and have become an integral part of modern people's lives [1] (i.e. search-and-rescue [2], forest and wildlife monitoring [3], post disaster assessment [4], law enforcement [5], border protection [6], engineering inspection [7], [8], large-scale mapping [9], etc). Compared to a helicopter of similar size, a quadcopter drone has many advantages [10]. Firstly, the system is mechanically simpler due to the fixed pitch configuration of its blades. Thus, it does not need swash plates to constantly incline the position of the rotating blade. This substantially reduces the complexity in both design and maintenance of the system. Since the drone employs four motors, the diameter of each propeller can be made significantly smaller for the same amount of the overall thrust. This also suggests lower kinetic energy, making the system safer for interactions with humans. In addition, it can offer lower operational noise and longer lifetime as its mechanical parts have lower frictional losses.

As a part of its telemetry system, the Parrot AR.Drone2 quadcopter (see Fig 1) is equipped with two cameras: a forward looking and a downward looking one; in addition to an ultrasound telemetry sensor, located below the central hull, for altitude measurements. The system is equipped with

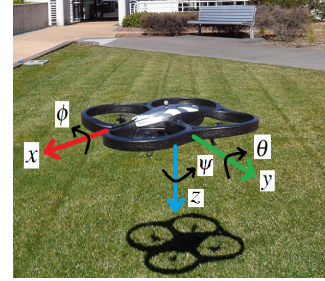


Fig. 1. The AR.Drone Quadcopter drone hovers in an outdoor environment. The notations of $[\phi \ \theta \ \psi]^T$ indicate the attitude of the drone that can give the rotation of the rigid body along the $[x \ y \ z]^T$ directions.

a six degree-of-freedom micro-electro-mechanical systems (MEMS)-based inertial measurement unit (IMU) to stabilise the pitch, roll, and yaw and to assist with the tilting control. Using its downward facing camera, the AR.Drone quadcopter monitors optical flow to maintain zero velocity during hover and a commanded velocity in other modes. The information from optical flow is very useful to minimise the drift from accelerometer and gyro, making it pivotal in stable operation of the AR.Drone. This is because the measurement data collected from the INS is noisy and drifting. Thus, the system is also very sensitive to the lighting condition of the environments.

When it comes to modelling the dynamics of the quadcopter drone, there are several challenges relating to the development of high fidelity dynamic models. Firstly, as an underactuated system, the dynamics of the quadcopter drone are highly non-linear, coupled with cross-coupling (also known as the *secondary effect*) between lateral and longitudinal loops with respect to the vertical loops. The system may also suffer from gyroscopic effects, if it rotates too quickly; in addition to having ground effect, if it flies too close to the ground level. Thus, to achieve high fidelity dynamic model, we will look at a non-linear system identification technique that has many advantages over the linear counterparts.

Although modelling complex dynamical systems from first principles can give an accurate understanding of their behaviours, this method is impractical in most cases because of the complexity and the unknown dynamic parameters of the systems. To overcome this issue, many researchers have proposed the use of some experiment-based identification

techniques, either gray box or black box methods, where no prior model is available, (e.g. ARX and ARMAX) to derive the linear mathematical model of the system. Although these methods seem to be easier to be implemented, they have some limitations; especially when faced by highly non-linear behaviours, in addition to limitations in the face of other uncertainties as well as the ill-defined condition of the modelled systems.

Current research papers in the literature still heavily focus on the traditional linear modelling techniques e.g. [11], [12], [13], [14], [15] and [16] as well as modelling from first principle [17] and [18]. Accordingly, to overcome the drawbacks of the conventional modelling techniques, we propose the development of high-fidelity non-linear modelling technique, inspired by human reasoning processes by means of the adaptive neuro-fuzzy inference system (ANFIS) identification method. This method has multiple benefits due to combined advantages between the Fuzzy Inference System (FIS) and the neural networks (NN) system.

The contribution of this paper, addressing current research gaps, can be elaborated as follows. Firstly, based on the AR.Drone2 transfer function models derived in [10], as an initial thrust of this research, we derive the ANFIS model of the lateral, longitudinal, and vertical dynamics of the system. Secondly, we benchmark the accuracy of our ANFIS model with respect to the accuracy of the MIMO transfer function obtained from linear system identification techniques. This research serves as a preliminary study towards the development of robust and intelligent autopilots.

The remaining of this paper is organised as follows. Firstly, Section II depicts the dynamics of the AR.drone2 quadcopter, including modelling from first principles and its MIMO transfer function depicted in [10], as the preliminary study of this research. Moreover, while Section III discusses the ANFIS architecture, Section IV describes the outcome of our ANFIS modelling, including its validation. Lastly, Section V concludes this paper.

II. SYSTEM DYNAMICS

This section discusses the dynamics of the AR.Drone2 quadcopter. To get a comprehensive understanding of its dynamics, we begin with the study of its mathematical models, derived from first principle.

1) *Modelling from First Principle:* According to the principles of the Newton-Euler's law, the differential equations, governing the translational dynamics of the rigid body of the AR.Drone2 quadcopter can be depicted as follows [19], [10]:

$$m \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} \sum k_F \omega_i^2 (\cos \psi \sin \theta + \cos \theta \sin \phi \sin \psi) \\ \sum k_F \omega_i^2 (\sin \psi \sin \theta - \cos \psi \cos \theta \sin \phi) \\ \sum k_F \omega_i^2 (\cos \phi \cos \theta) - mg \end{bmatrix} \quad (1)$$

where m denotes the mass of the system, $(\ddot{x}, \ddot{y}, \ddot{z})$ represent the second derivative of (x, y, z) position, $[\phi, \theta, \psi]^T$ indicate the attitude (roll, pitch and yaw) of the system, k_F depicts the motor and force constant, which are assumed to be identical,

and ω_i indicates the angular velocity of i -motor, and g is the gravity constant.

Furthermore, the rotational dynamics of the system can be presented as follows [10], [19]:

$$\begin{bmatrix} I_{xx} \dot{p} \\ I_{yy} \dot{q} \\ I_{zz} \dot{r} \end{bmatrix} = \begin{bmatrix} l k_F (\omega_2^2 - \omega_4^2) \\ l k_F (\omega_3^2 - \omega_1^2) \\ k_M (\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \end{bmatrix}, \quad (2)$$

where I_{xx} , I_{yy} , and I_{zz} denotes second moment, which represents the rational inertia along (x, y, z) -axis and l represents the distance between the motor to the centre of the quadcopter, $[p, q, r]^T$ indicates vectors of the angular velocity of the body frame of the aircraft (rotational dynamics).

It should be pointed out that the mathematical models in (1) and (2) are derived under several assumptions. Firstly, we assume that the body and the blades are rigid. Second, the amount of the aerodynamics drag is sufficiently small, given the vehicle travels slowly. The gyroscopic torque is assumed to be negligible, given the vehicle also rotates slowly. In addition, we also assume there is no ground effect, provided the altitude of the vehicle is far higher than the ground level. Also, we assume that the system is symmetrical along its lateral and longitudinal axes.

Furthermore, the relationship between the input angular velocities with respect to the attitudes of the drones and its vertical motions are given in (3)-(6) as in [10]:

$$\Delta \omega_\phi = \frac{1}{2} (\omega_2 - \omega_4), \quad (3)$$

$$\Delta \omega_\theta = \frac{1}{2} (\omega_3 - \omega_1), \quad (4)$$

$$\Delta \omega_\psi = \frac{1}{4} (\omega_1 - \omega_2 + \omega_3 - \omega_4), \quad (5)$$

$$\Delta \omega_F + \omega_h = \frac{1}{4} (\omega_1 + \omega_2 + \omega_3 + \omega_4), \quad (6)$$

where $\omega_h \doteq \bar{\omega}_1 = \bar{\omega}_2 = \bar{\omega}_3 = \bar{\omega}_4 = \sqrt{\frac{mg}{4k_F}}$, $\bar{\omega}_{1,2,3,4}$ indicates an ensemble average of the angular velocity of each motor. Moreover, $\Delta \omega_F$, $\Delta \omega_\phi$, $\Delta \omega_\theta$, $\Delta \omega_\psi$ denote the angular velocity of the four inputs, namely, ones creating the vertical motion, roll, pitch, and yaw, respectively.

2) *Multi-Input, Multi-Output (MIMO) transfer function:* In this section, we will recall the multi-input, multi-output (MIMO) transfer functions of the longitudinal, lateral, and vertical loops, derived from linear system identification technique, as initially studied in [10]. This transfer function can be regarded as the linearised model of the system about the hover point with attitudes less than 30 deg to guarantee its accuracy. The overall dynamics of the system can be represented using the block diagram as illustrated in Fig 2.

Firstly, the transfer function of the pitch (longitudinal) loop can be depicted as follows [10]:

$$G_x(s) = \frac{v_x(s)}{\theta_{in}(s)} = \frac{0.7995}{s^2 + 4.1881s + 7.1429}. \quad (7)$$

The transfer function in (7) has complex conjugate poles at $s_{1,2} = -2.0941 \pm 1.6607i$ that give a damping factor $\zeta = 0.784$

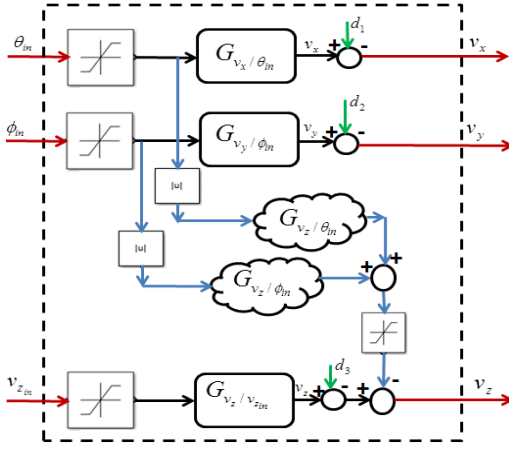


Fig. 2. Non linear model of the inner loop (attitude) dynamics: control input vector is given by $[\theta_{in} \ \phi_{in} \ v_{zin}]^T$, while the output controlled variables are $[v_x \ v_y \ v_z]^T$ and the external disturbance vector (e.g. wind gusts) is represented by $[d_1 \ d_2 \ d_3]^T$.

and undamped natural frequency $\omega_n = 2.67\text{rad/s}$. The transfer function of the roll (lateral) loop is given as follows [10]:

$$G_y(s) = \frac{v_y(s)}{\phi_{in}(s)} = \frac{116.24s + 67.2488}{1000s^2 + 982.664s + 301.338}. \quad (8)$$

The transfer function in (8) has complex conjugate poles at $s_{1,2} = -0.491 \pm 0.245i$ that give a damping ratio $\zeta = 0.895$ and undamped natural frequency $\omega_n = 0.549\text{rad/s}$. The transfer function of the vertical loop is given by [10]:

$$G_z(s) = \frac{v_z(s)}{v_{zin}(s)} = \frac{17.6902s^2 + 37.7142s + 422.62}{s^4 + 11.6091s^3 + 68.394s^2 + 257s + 688.244}. \quad (9)$$

The transfer function in (9) has two pairs of complex conjugate poles at $s_{1,2} = -5.1 \pm 2.73i$ that give $\zeta = 0.882$ and $\omega_n = 5.78\text{rad/s}$ as well as $s_{3,4} = -0.705 \pm 4.48i$ with $\zeta = 0.155$ and $\omega_n = 4.54\text{rad/s}$.

As illustrated in Fig. 2, there is an undesirable effect (also known as secondary effects) between the lateral and the longitudinal motions with respect to the vertical (z) loop, given by $\frac{G_{vz}}{G_{\theta_{in}}}$ and $\frac{G_{vz}}{G_{\phi_{in}}}$. This results in the loss of altitude during the maneuver of the drone, which occurs due to a reduced vertical component of thrust when the system is at non-zero attitude. Also, the vertical drag acting on the body of the drone due to the rotor downwash will tend to cause negative vertical velocity.

III. ANFIS ARCHITECTURE

ANFIS is a powerful tool to model the dynamic behaviors of non-linear systems, such as in the case of our quadcopter drone. The system leverages the benefits of both Fuzzy Inference Systems (FIS) that can model the qualitative aspects of human reasoning process and the Neural Networks (NN) systems that has the ability to learn and to recognise patterns in the data and to develop their relationship [20].

Combining the advantages of the fuzzy systems and the neural networks, ANFIS can be regarded as an adaptive-

network-based fuzzy inference systems. Utilising the input-output data, ANFIS constructs the dynamic model of the system based on the human knowledge, manifested in the ‘If-Then’ FIS rules, supported by its hybrid learning algorithms to model the expert behaviours [21].

The system utilizes fuzzy modelling technique to learn about the data set information. It determines the parameters of the fuzzy membership functions to allow the FIS to model the dynamic behaviours of the system. Thus, the main task of the learning process is to optimise the FIS parameters to minimise the training error. In what follows, we will highlight two major components of the ANFIS, namely, the fuzzy inference systems and the neural networks.

A. Fuzzy Inference Systems

We employ Sugeno-based Fuzzy Inference System (FIS) to model the performance of our quadcopter dynamics. The first important step in FIS is fuzzification, aiming to transform crisp input, defined as a set of measurement data obtained from sensors (e.g. the attitude of the aircraft); into the membership linguistic terms (fuzzy sets). Another important step is the defuzzification process, which is related to the process of transforming fuzzy grade into output signals e.g. the velocity components of the aircraft. There are many defuzzification methods in the literature [22], [23] (e.g. true center of gravity, fast centre of gravity, fuzzy mean, mean of maxima, sparus, plateau average, capitis, and clivusos). In this paper, nonetheless, we employ the centroid method, which implements the true centre of gravity to determine the crisp output signal.

B. Neural Networks

In ANFIS, we use the neural networks to improve the accuracy of the FIS system. Leveraging on the benefits of its learning capability, NN can be regarded as an adaptive feedforward system, comprising of the interconnected neurons to exchange the messages, as represented by $y(k+1) = f_n(y(k), u(k))$, where f_n is the neural function. Its numerical weights can be easily tuned based on the learning experience of the system, making it suitable for highly-parallel and intelligent learning as in the case of our quadcopter drone. This leads to its ability to deal with robustness with respect to disturbances.

1) *Back Propagation of Errors*: In this section, we will discuss the most widely implemented training method in the artificial neural networks, known as the “back propagation technique”. Given a set of input-output data pairs, namely, $\mathbf{X} = [x_1, \dots, x_N]^T$, $\mathbf{y}^* = [y_1^*, \dots, y_N^*]^T$, the error signal e_k is defined as the difference between the desired output y^* with respect to the output of the network y , namely, $e_k = y_k^* - y_k$. This error signal will be used to adjust the weights in the NN, so that the quadratic function in (10) will be minimised [24]:

$$J = \frac{1}{2} \sum_{k=1}^N e_k^2. \quad (10)$$

Minimising (10), the update rule in (11) will be applied:

$$\mathbf{w}(n+1) = \mathbf{w}(n) - \alpha(n) \nabla J(\mathbf{w}(n)), \quad (11)$$

where $\mathbf{w}(n)$ is the weight at n -iteration, $\alpha(n)$ is the parameter of the learning rate, and $\nabla J(\mathbf{w})$ denotes the Jacobian vector, which can be mathematically given as:

$$\nabla J(\mathbf{w}) = \left[\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}_1}, \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}_2}, \dots, \frac{\partial J(\mathbf{w})}{\partial \mathbf{w}_M} \right]. \quad (12)$$

Accordingly, the non-linear optimisation in (12) can be solved using the second order gradient methods as follows:

$$J(\mathbf{w}) \approx J(\mathbf{w}_0) + \nabla J(\mathbf{w}_0)^T (\mathbf{w} - \mathbf{w}_0) + \frac{1}{2} (\mathbf{w} - \mathbf{w}_0)^T \mathbf{H}(\mathbf{w}_0) (\mathbf{w} - \mathbf{w}_0), \quad (13)$$

where $\mathbf{H}(\mathbf{w}_0)$ is the Hessian. After a few derivations, the update rule in (13) will become (14):

$$\mathbf{w}(n+1) = \mathbf{w}(n) - \mathbf{H}^{-1}(\mathbf{w}(n)) \nabla J(\mathbf{w}(n)). \quad (14)$$

C. ANFIS Mathematical Model

Combining the concept of FIS and NN, ANFIS consists of several parts e.g. the *antecedent* and *conclusion* parts connected by networked fuzzy rules [24], [25]. The first layer performs fuzzification, as described by $O_i^1 = \mu_{A_i}(x)$, where A_i indicates the fuzzy linguistic variables associated with each node. For instance, for typical a bell shape linguistic function, its mathematical model can be given as follows [24]:

$$\mu_{A_i}(x_j, c_{ij}, \sigma_{ij}) = \exp \left\{ -\frac{(x_j - c_{ij})^2}{2\sigma_{ij}^2} \right\}, \quad (15)$$

where c_{ij} and σ_{ij} are the membership parameters, and x_j denotes the input variables. The second layer performs multiplications (the fuzzy AND of the antecedent), before sending the product out. Meanwhile, the third layer normalises the MFs as follows: $\bar{w}_i = \frac{w_i}{\sum_i w_i}$, and the fourth layer concludes the fuzzy rules by computing the following function $O_i^4 = \bar{w}_i f_i$. Finally, the last layer computes the output of the fuzzy system by summing up the output of the fourth layer:

$$\sum_i \bar{w}_i f_i = \frac{\sum_i w_i f_i}{\sum_i w_i}. \quad (16)$$

This process is also known as defuzzification. For the first order TS model, the input-output equation is given as follows:

$$y = \sum_{i=1}^K \gamma_i(\mathbf{x}) (\mathbf{a}_i^T \mathbf{x} + b_i), \quad (17)$$

where

$$\gamma_i(\mathbf{x}) = \frac{\prod_{j=1}^p \exp \left\{ -\frac{(x_j - c_{ij})^2}{2\sigma_{ij}^2} \right\}}{\sum_{i=1}^K \prod_{j=1}^p \exp \left\{ -\frac{(x_j - c_{ij})^2}{2\sigma_{ij}^2} \right\}}. \quad (18)$$

Following the gradient-based learning, the consequent parameters of b_i , c_{ij} , and σ_{ij} can be determined using the back-propagation technique as follows:

$$\frac{\partial J}{\partial b_i} = \frac{\partial J}{\partial e} \cdot \frac{\partial e}{\partial b_i} = -\gamma_i e, \quad i = 1, 2, \dots, K. \quad (19)$$

given the following *update* law:

$$b_i(n+1) = b_i(n) + \alpha(n) \gamma_i e, \quad i = 1, 2, \dots, K. \quad (20)$$

Likewise, for the centres c_{ij} and the spread σ_{ij} of the Gaussian function in (15), the update rules in (21) are applied:

$$\begin{aligned} c_{ij}(n+1) &= c_{ij}(n) + 2\alpha(n) \gamma_i e [b_i - y] \frac{x_j - c_{ij}}{\sigma_{ij}^2}, \quad i, j = 1, 2, \dots \\ \sigma_{ij}(n+1) &= c_{ij}(n) + 2\alpha(n) \gamma_i e [b_i - y] \frac{(x_j - c_{ij})^2}{\sigma_{ij}^3}, \quad i, j = 1, 2, \dots \end{aligned} \quad (21)$$

IV. ANFIS IDENTIFICATION

This section discusses the modelling process and the outcome of our adaptive neuro-fuzzy system identification technique.

A. Dryden Wind Turbulence Model

To begin with, we will discuss the well-known Dryden wind turbulence models in (22), (23), (24), widely implemented by the United States Department of Defence, to model the input identification signal in real flight environments [26], [27]). This model treats continuous gusts as a *spatially varying stochastic* process.

Since the attitude of an aircraft is *directly affected* by wind gusts, we can employ this model to generate the input identification signal for our drone. Given the three linear velocity components of the gusts, namely, u_x , u_y , u_z , the characteristics of the wind gusts along x , y , z axes can be modelled as follows. Firstly, the Dryden gusts along x axis can be represented by a first order transfer function as follows:

$$G_{w_x}(s) = \sigma_u \sqrt{\frac{2L_u}{\pi v}} \frac{1}{1 + \frac{L_u}{v}s}, \quad (22)$$

where σ_u and L_u indicate the intensity and the scale length of the turbulence along x axis while v depicts the speed in which the aircraft moves through the gust field.

Next, the characteristics of wind gusts along y axis is given in the form of second order transfer function as follows:

$$G_{w_y}(s) = \sigma_v \sqrt{\frac{2L_v}{\pi v}} \frac{1 + \frac{2\sqrt{3}L_v}{v}s}{(1 + \frac{2L_v}{v}s)^2}, \quad (23)$$

where σ_v and L_v indicate the intensity and the scale length of the turbulence along y axis.

Lastly, the characteristics of wind gusts along z (vertical) axis can be represented using the following second order transfer function:

$$G_{w_z}(s) = \sigma_w \sqrt{\frac{2L_w}{\pi v}} \frac{1 + \frac{2\sqrt{3}L_w}{v}s}{(1 + \frac{2L_w}{v}s)^2}, \quad (24)$$

where σ_w and L_w indicate the intensity and the scale length of the turbulence along z (vertical) axis.

B. Modelling Outcome

In this section, we benchmark the efficacy of our ANFIS model with respect to the performance of the linear transfer

function as a preliminary study of this research (7), (8), and (9), obtained by linear system identification techniques. In what follows, we will present the outcome of our ANFIS model for three different loops (longitudinal, lateral, and vertical).

1) *Longitudinal Loop*: To achieve a delicate balance between the accuracy and the computational trade-off, we employ seven Gaussian membership functions for our FIS as illustrated in Fig. 3 first row. The resulting structure of the FIS ‘if then’ rules, for all three loops follows the Sugeno’s rule as described in Tables I, II, III, respectively. Compared

TABLE I
ANFIS ‘If-Then’ RULES FOR THE LONGITUDINAL LOOP

1.	If input is in-mf1 then $y(k) = 0.05926u - 0.008239$
2.	If input is in-mf2 then $y(k) = 0.07308u - 0.01798$
3.	If input is in-mf3 then $y(k) = 0.0531u - 0.01873$
4.	If input is in-mf4 then $y(k) = 0.05685u - 0.0109$
5.	If input is in-mf5 then $y(k) = 0.04775u + 0.02235$
6.	If input is in-mf6 then $y(k) = 0.06085u + 0.005249$
7.	If input is in-mf7 then $y(k) = 0.05026u + 0.01376$

to the Mamdani fuzzy, the TS fuzzy is more computationally efficient, making it suitable for modelling, optimisation, and adaptive tasks; in addition to its ability to guarantee the continuity of the output surface.

The optimised FIS membership function for the pitch loop is given in Fig. 3 first row (left). While the training error for 50 epochs training interval is given in the second row left, the input-output relation is given in the second row right of Fig. 3. Meanwhile, the modelling and validation signal are given in the third row.

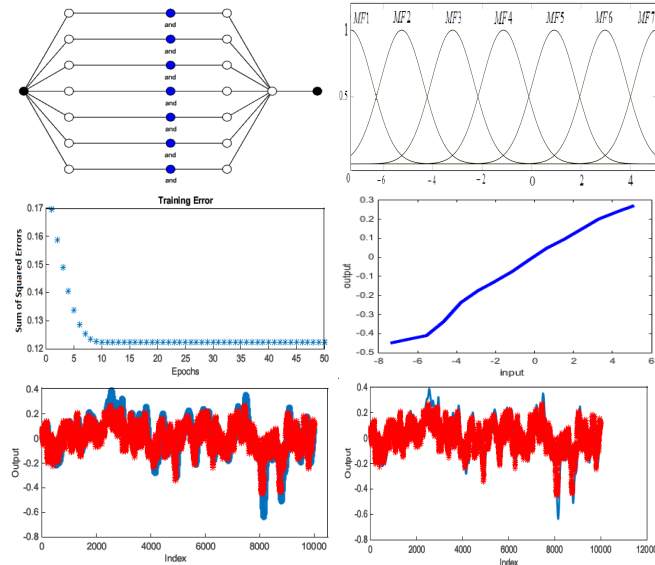


Fig. 3. ANFIS architecture for the longitudinal (pitch) loop: (a) First row left: The structure of the neural network (7 neurons), (b) First row right: the resulting FIS linguistic function, (c) Second row left: training error, (d) Second row right: fuzzy I/O relation, (e) Third row left: modelling outcome (ANFIS vs. transfer function), (f) Third row right: validation signal.

2) *Lateral Loop*: Likewise, for our lateral loop, while still employing the same structure of neural networks as in Fig. 4 first row left, the optimised FIS membership function used to model the dynamics of the roll loop is given in Fig. 4 first row right. Furthermore, the convergence of the training error is given in Fig. 4 second row left, and the input-output FIS mapping is highlighted in Fig. 4 second row right. Lastly, while the training signal is given in Fig. 4 last row left, the validation signal is illustrated in Fig. 4 last row right.

TABLE II
ANFIS ‘If-Then’ RULES FOR THE LATERAL LOOP

1.	If input is in-mf1 then $y(k) = 0.06315u - 0.006975$
2.	If input is in-mf2 then $y(k) = 0.1302u - 0.1563$
3.	If input is in-mf3 then $y(k) = 0.171u + 0.01207$
4.	If input is in-mf4 then $y(k) = 0.02446u - 0.05691$
5.	If input is in-mf5 then $y(k) = 0.07466u + 0.1014$
6.	If input is in-mf6 then $y(k) = 0.1842u - 0.0001937$
7.	If input is in-mf7 then $y(k) = 0.1358u + 0.0663$

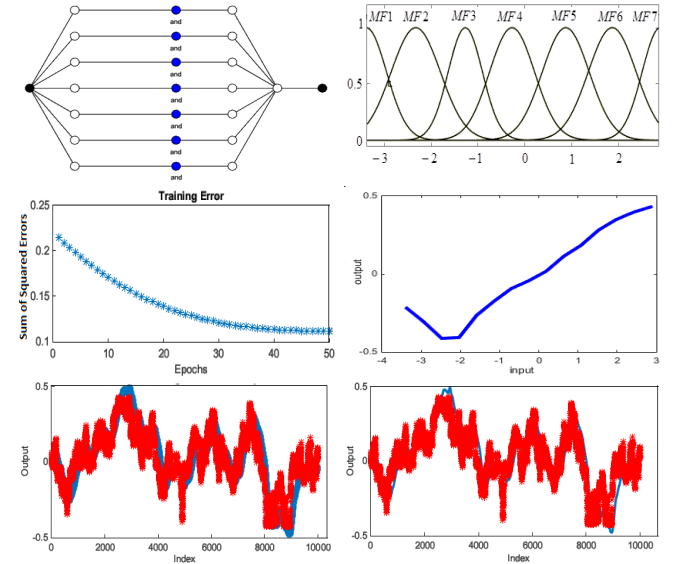


Fig. 4. ANFIS architecture for the lateral (roll) loop: (a) First row left: The structure of the neural network, (b) First row right: the resulting FIS linguistic function, (c) Second row left: training error, (d) Second row right: fuzzy I/O relation, (e) Third row left: modelling outcome (ANFIS vs. transfer function), (f) Third row right: validation signal.

3) *Vertical Loop*: The vertical loop also employs the same neural structure with seven membership function as highlighted in Fig. 5 first row. However, compared to both the longitudinal and the lateral loops, the resulting FIS membership function of the vertical loop is less symmetrical due to its highly non-linear nature (e.g. secondary effects) as discussed in Section II.

V. CONCLUSION

Leveraging on the benefits of soft computing technique, we have derived the ANFIS model of the AR.Drone quadcopter.

TABLE III
ANFIS 'If-Then' RULES FOR THE VERTICAL LOOP

1.	If input is in-mf1 then $y(k) = 0.6367u + 0.0888$
2.	If input is in-mf2 then $y(k) = 0.269u - 0.3559$
3.	If input is in-mf3 then $y(k) = 0.5926u + 0.0319$
4.	If input is in-mf4 then $y(k) = 0.8837u - 0.009114$
5.	If input is in-mf5 then $y(k) = 1.173u - 0.2579$
6.	If input is in-mf6 then $y(k) = 0.8607u - 0.5853$
7.	If input is in-mf7 then $y(k) = 0.6749u - 0.1566$

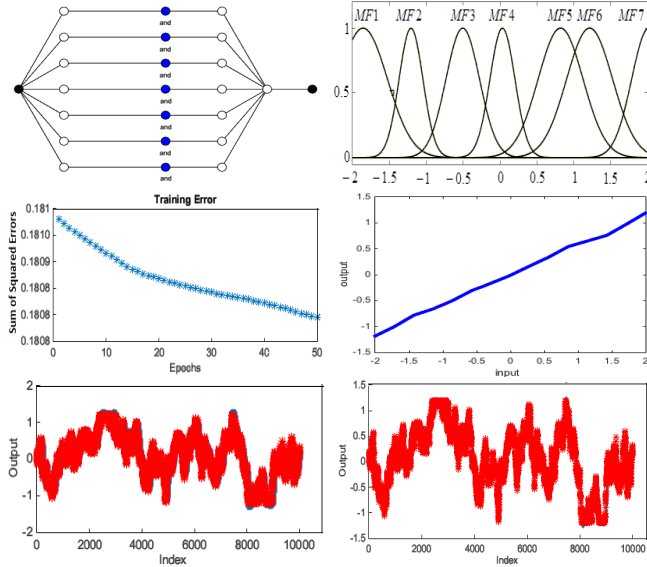


Fig. 5. ANFIS architecture for the vertical loop: (a) First row left: The structure of the neural network, (b) First row right: the resulting FIS linguistic function, (c) Second row left: training error, (d) Second row right: fuzzy I/O relation, (e) Third row right: modelling outcome (ANFIS vs. transfer function), (f) Third row right: validation signal.

We have demonstrated the accuracy of the proposed ANFIS model through extensive validation process. In the absence of a complex mathematical model, the superiority of the proposed ANFIS model is related to the combined benefits of a neural network technique, which has the capability of learning, and the superiority of the FIS, which relies on the expert knowledge, making it more transparent. The proposed model is also highly suitable to capture the non-linear dynamics of the quadcopter drone.

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